

Team 02

Project 1: Galaxy Trucker Quick Reference Guide

IE 5760

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Introduction

Galaxy Trucker (Chvatil, 2007) is a semi-competitive space-themed board game in which players rapidly construct a spaceship from modular tiles of varying types and then fly their ship through a course of obstacles to gather and sell cargo. Anyone who makes more money than they lose wins, but some players might win more than others.

The rulebook of this game was chosen as a candidate for human factors improvement because of its apparent gaudiness and incoherence to the authors. User testing was conducted to identify user needs from the Galaxy Trucker rulebook, and affinity diagramming based on user testing notes revealed visual overload, unclear goals, high short term memory load, and graphic design layout issues to be the biggest obstacles to users feeling comfortable and confident in playing the game.

To narrow the scope, we focused on the needs of returning users who might need to revisit the rulebook to remind them of certain rules and game flow. Quick reference guides are often used to aid these users (Björkman, 2019) and use icons and hierarchical structure to aid construction of a top-down guided salience map for visual search (Nothdurft, 2006). We created a prototype quick reference guide based on these principles and had we more time, we would test this against the standalone rulebook to verify that it leads to better rule retention and user satisfaction. This report serves as a summary of user testing, justification for redesign, and discussion of the impact of rulebook design choices on user experience.

Background & Justification

The instruction layout for a board game plays an important role in shaping how a player interprets and feels about the game. When the game pieces and procedure of the game are clear and organized, it can improve player performance, reduce errors, and enhance player satisfaction. A task-centered design of a manual (or, in this case, board game instructions) can help users overcome issues that arise (Lewis & Reiman, 1994). This can be done by “speaking the users’ language,” maintaining brevity and keeping the instructions “lean” by including only material that is relevant. Game instructions should be designed with the user’s goals in mind, not just based on the structure of the game itself.

When a player is returning to a game, they rely on long-term memory to recall their previous experiences and use their working memory to process instructions. If players are presented with too much information or information is poorly organized, the ability to process might decline. With player working memory, “chunking benefits the operations in working memory in several ways” including increasing capacity, aiding retention through meaningful associations, and easily rehearsing items to transfer to long-term memory (Wickens et al., 1998; Norman, 1988). If the game instructions are too complex or not well organized, then players may struggle to process and apply them.

The quick reference board game instructions must communicate through design. Crilly et al. (2004) describes how emotional responses are shaped by both instrumental and aesthetic factors. “Instrumental emotions (such as disappointment or satisfaction) stem from perceptions of whether a product will assist the user in achieving their objectives. Aesthetic emotions (such as disgust or attraction) relate to the potential for products to ‘delight and offend our senses’”. For returning players, the instructions should support the player’s objectives and provide a pleasing experience that encourages re-engagement.

The 2007 Galaxy Trucker instructions appear complex, and user feedback suggests needs for improvement to meet the needs of both new and returning users. The overload of information, graphics, and game playing is why this game is a good contender for instruction redesign. On Reddit forums, players have reported their frustrations:

- *“What I could tell from the reviews it was fairly lighthearted, not too complex and a lot of fun... I'm a bit disappointed. I'd like to hope this will get better on replays.”*
- *“It sounds like you're not actually playing the game right?”*
- *“The family and I were going through the learner game and following the book, but it wasn't clear at all how we move forward.”*
- *“Please help me, I really don't understand this point in the rules.”*
- *“So I just got the game and I think were [sic] playing it wrong?”*
- *“I can't seem to find the rule how the dice work in this game (maybe I'm just blind)”*
- *“Galaxy Trucker is such a hit or miss game it seems. 2 of my friends adore it the game and another two of us would be perfectly fine never playing it again”*

These responses highlight two things: users are struggling to understand the rules of the game based on the rulebook, and they assume they must be the problem because the issues they are encountering seem simple. This follows a common pattern indicating bad design, much like the poor door-openers who think themselves foolish for pushing on a badly designed “pull” door (Norman, 1988). This suggests a need for a rulebook with which it is faster to find a rule than to ask about that rule on Reddit.

Returning player quick-reference guides need to meet and align with player needs and make it easier to come back to the game without re-reading the full instructions. Experienced players don’t need comprehensive information, but rather streamlined overviews. “The limitations can be broken down into two factors: how much information is presented to the operator and in what way the information is presented. First, designers must be careful not to present a large quantity of information and expect people to be able to process it in a short time” (Wickens et al., 1998).

By narrowing the scope, it follows Norman’s usability principle of visibility. Norman states “usability and learnability are improved when the user can easily see what commands and options are available” (Matz, 2012). For Galaxy Trucker, this means providing visuals for game

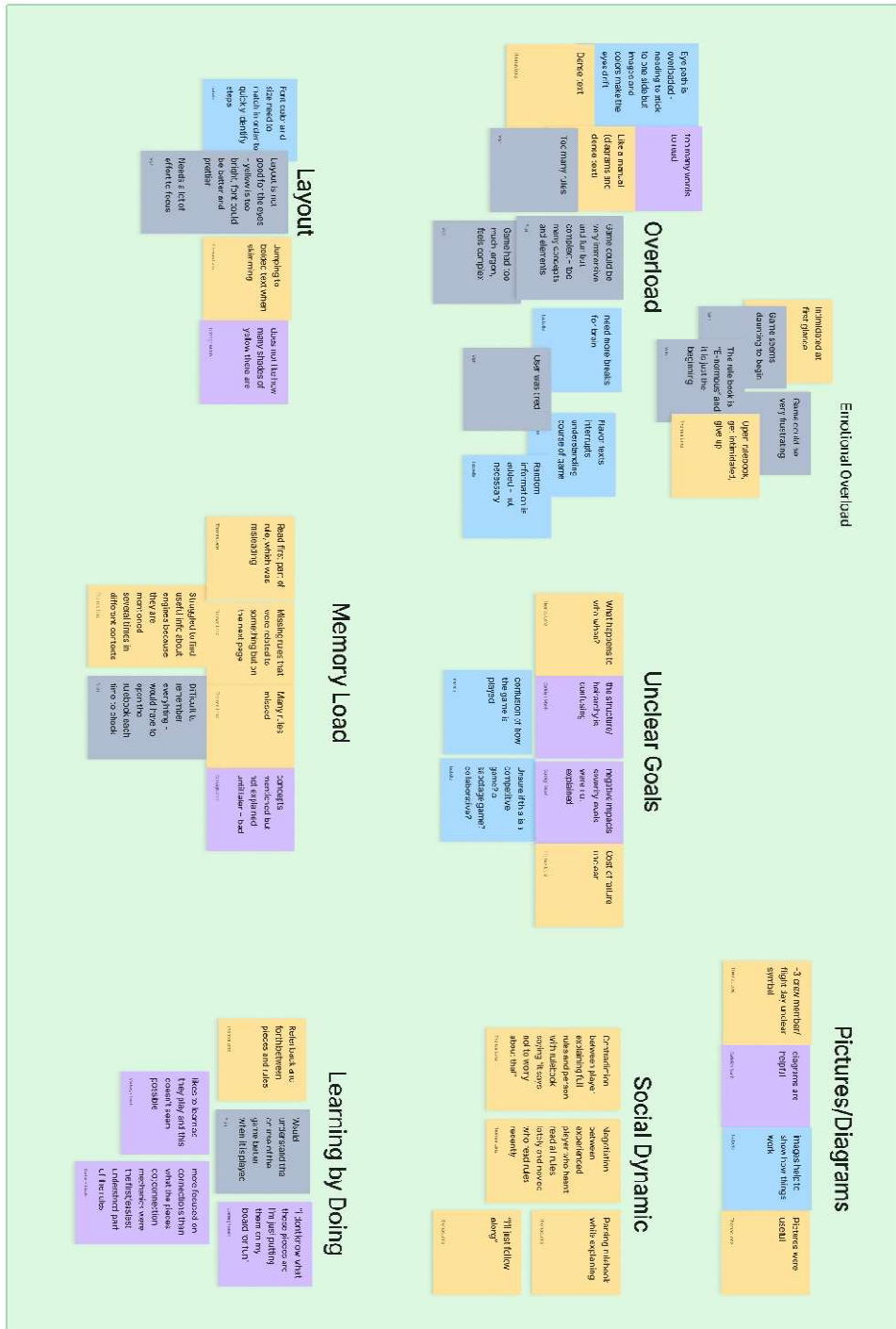
components and a reference to each phase of the game. In conclusion, a clear and organized layout for a quick reference guide will improve returning player performance, reduce errors, and enhance satisfaction.

User Testing

Participants for the experiment or testing were first-time users/players of Galaxy Trucker. This was to gather perspectives from both the groups of people who were interacting with the rule book for the first time. The material used for this study was the 2007 Galaxy Trucker rulebook.

The tasks for the initial user testing to understand the pain points were to go through the first 5 pages of the rulebook, which described how one builds their first ship. They were then asked to interpret and relay their inferences back to the experimenter. Five participants were asked to relay and set up the game and answer any clarifications related to the game. Based on the data gathered, prominent themes were identified and grouped through affinity mapping – namely, overload especially, emotional overload, Unclear Goals, Layout, Memory Load, Learning by doing, Social Dynamic and the use of pictures/diagrams. Once the pain points and user needs were understood, the key areas for improvement were decided. These were mainly reducing the cognitive load on the players and facilitating learning by improving the layout and simplifying the rules to reduce jargon. The redesign of the rulebook was ideated and designed, and the targeted audience for the redesigned rulebook was returning users of the game, as we wanted to test it on the same participants.

Fig. 1 Affinity Map of information gathered from the User Testing.



Redesign Approach

Literature Findings

The redesign of the rule book was influenced and supported by previous literature. The wide amount of information and rules presented in Galaxy Truckers (2007) version was broken down into chunks of shorter and more easy-to-understand format. Chunking was found to be helpful with symbolic sequence learning and reducing cognitive load (Solopchuk et al., 2016).

Progressive disclosure was used to properly sequence information in the rulebook across pages and make the rulebook easier to learn and reduce the number of errors (Spillers et al., 2015).

More aesthetically pleasing pictures and icons were used to reduce cognitive load, increase retention and to make learning the game easier (Zu et al., 2018).

The redesign of the rulebook also incorporated the use of examples to enhance example-based learning to reduce the cognitive load, remove any ambiguity in the rules, and help with their retention of the rules, especially for first-time users (Chen et al., 2023). The tabs at the right-hand corner of the page and the positioning of the images or pictures close to the relevant rules also help increase their contextual salience, which has been found to increase the speed of target detection in a visual search task (Northdurft, 1993). They also help with the top-down processing (Wickens et al., 1998) of the rules, which ultimately increases the efficiency of understanding the rules and relaying them to play the board game. Additionally, the F-shaped pattern for organizing the text in the helpsheets helps the players read the text in a way that facilitates natural mapping, i.e., scanning from left to right along the top and then scanning the left side of the page to look for visual cues (Soegaard, 2021).

Redesign

Based on player feedback on Reddit and initial user testing, the results showed the rulebook was overwhelming. Users struggled to recall component pieces, found the layout of the phases buried amongst “flavor text”, and a large amount of time understanding setup steps. These pain points pointed to three core needs for returning players:

1. Rapid recall of component functions
2. Clear visibility of game phases
3. Minimal navigation through text

Analyzing player insights with theory produced a quick-reference guide that chunks text and defines sections with corresponding titles, groups components with their title and an excerpt of their function or effect and lays out the game phases.

Table 1. Redesign Rationale

Improvement	Rationale	Secondary Benefit
Chunked section titles	Reduces working-memory load by organizing content (Wickens et al, 1998)	Improves scannability
Icon and text visuals	Combines icons and text into cohesive units (Zu et al, 2018)	Increases comprehension and retention
Flow of game phases	Minimizes cognitive load by laying out game procedure (Wickens et al, 1998)	Speeds up recall, reduces setup time
Hazard and scoring	Progressive disclosure is used to separate information types (Spillers 2015)	Less friction amongst users when it comes to scoring
Lean task content	Keeps instructions focused on returning player's goals, eliminating irrelevant information (Lewis & Reiman, 1994)	Prevents tutorial fatigue
High-contrast typography and white space	Improves aesthetic engagement and readability through clear layout (Crilly et al, 2004 & Matz, 2012)	Meets accessibility standards (Color: AAA)

(See Appendix for more designs)

Galaxy Trucker 2007 Game Instructions



Quick-Reference Guide



Discussion

Due to time constraints, the team decided to focus on assisting returning Galaxy Trucker players rather than redesigning the entire rulebook. This help sheet is designed for quick reference throughout the course of the game and incorporates both user feedback and theory to aid in understanding. The information is organized with a heading-subheading structure, so users can scan the pages quickly to reach the section they need. Additionally, there are title tabs in the top-left corner of the pages, allowing players to flip through and find the correct section without fully opening the document to each page. Once the desired page is reached, colored images of the physical game pieces catch user's attention to aid in the visual search. The plain white background, while somewhat less thematic than the original design, contrasts with the graphics and defines gaps between sections of text, encouraging visual chunking.

Although the help sheet was designed with returning players in mind, it will also be a valuable, time-saving resource to new players as they play their first game. One unintended benefit is the ability for two players to reference the rules at once, as the sheet will be used in tandem with the rulebook, not instead of it. This was a concern observed in user testing, where some players felt themselves to be inactive participants or wished they could see the rulebook while

someone else was using it. As the instructional documents can be in two places in the room, visual learners will also have more opportunity to interact with their preferred medium.

While this project is promising, it is also limited by multiple constraints, the largest of which is time. The original goal of redesigning the entire Galaxy Trucker rulebook was quickly deemed infeasible due to the two-week turnaround time from the project proposal. Despite the limited scope of the help sheet, it utilizes the same methods to improve the readability of instructions as a full redesign. Additionally, this design has not been through the iterative design cycle and has not been evaluated for colorblind accessibility. Both of those processes are high on the priority list for any future work done on this project.

Future Work

Future improvements made to the Galaxy Trucker rulebook can be built on the simple style used in the quick reference guide and should be applied across the entire manual. This largely consists of rewriting the advances and in-depth sections with the same clear language, examples and visuals, so players are not confused or overwhelmed. This rewrite should be tested with both new and returning players, to validate the claim that the layout supports both quick learning and works as a refresher.

An additional step is to arrange the rules in a way that matches how people understand information most effectively. A step-by-step explanation, short overviews and select relevant images can make the difficult parts easier to understand. The rules should not be categorized by simple and complex, but by some logical grouping, like game phase, to minimize short term memory load. These changes would make Galaxy Trucker more enjoyable and highlight how good instructional design can make even complex games more approachable.

Conclusion:

In this project, we have identified a board game rule book that poses a significant obstacle to user confidence and ability to play the game. Through user testing, we noted user challenges and categorized them through affinity diagramming. Visual overload, unclear goals, high short term memory load, and graphic design layout issues seemed to be the most common impediments to user success, based on user comments.

We posit that these issues arise from dense text that results in cognitive overload, paired with poor graphic design. To address these issues for a narrowed user group of returning users who need to be refreshed on rules, we developed a quick reference guide prototype that relies on symbols, F-shaped structure, and logical groupings to enable top-down processing and reduce the cognitive load associated with visual search. If more time was available, we would test this new reference guide with appropriate users to prove its efficacy, but at the very least we believe that it is useful. We do plan to print it for use with our own board game nights, so at least we are not violating Lewis and Rieman's suggestion to "never work on something if you

ought to be a user for it but you aren't" (Lewis and Rieman, 1994, p. 13). Furthermore, we do believe that it will lead to faster searches for rules and lower cognitive load for our target users, leading to easier and more fulfilling gameplay.

Author Contribution Statement

Thomas worked on the Introduction and Conclusion, and the rule specifics of the prototype. Isabelle worked on the Background and Redesign portion of the Redesign Approach, while Visruta worked on the Method and Literature Findings of the Redesign Approach. Carleigh worked on the Discussions section and the slide deck theme, and Heeral worked on the Future Works section. All members of the team contributed to the user testing and redesigning the rulebook on Figma.

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Appendix

USER TESTING NOTES

Participants 1-3

Task	Success Rate	P1	P2	P3
Task 1: Understand course of game	75.00%	1	1	1
Task 2: Build first ship	50.00%	1	0.5	0.5
Time taken Task 1		2 min	2 min	2 min
Time take Task 2		45 min	12 min	15 min
Likeliness to play		4/10	2/10	3/10
Notes Task 1		block of yellow on first page is interrupting understanding course of the game would prefer it off to the side, it is part of the story line, but it doesn't help to understand the intial run through of how to play the game separate round 1, 2 and 3 into different paragraphs too much text, needs to differentiate between sections make different rounds more apparent (in bold) font color and size need to match to be able to quickly identify the steps cosmic credits??? confident level: need to play to understand, 6/10	likes the black on white contrast of the text unsure if yellow block is necessary to read confused about mention of credit rating only one sentence below the yellow block looks bad formatting wise "why is this weird drawing here" doesn't like the angle cutoff of the main textbox multiple fonts look bad/confusing	Did not have a lot of trouble understanding the course of the game. Would better understand it if the game is played.
Notes Task 2		blurb under subheading needs to be reworded, confusing, a lot to read eyes first go to yellow block of text, thrown	confused start, figuring out how center piece connects. said the diagram helped "I don't know what these pieces are for I'm	Game has too much jargon, feels complex right from the beginning Seems daunting to begin

		into random story and it doesn't make sense eyes went to the right to build ships because reading left to right instead of building set up - don't know where to go to first eyes are jumping all around, understand 2 columns, text vs images, hard to read text - likes the use of colors but its too much confused with set up, need more information on which each piece is called rather than being thrown all of the components together (spaceship board, etc) like the diagram explaining connectors, different images helped to have more detail of how they connect consistency of the checked vs incorrect is off, the one above is checks then incorrect compared to the one below, one more thought change cabin section under components, reasoning why crew sizes is better, first time mentioning it - later on canons has a good explanation as to why you want more canons eyes are jumping back between columns, have to force self to read each column, less enthusiastic	just putting them on my board for fun" jumped into building the shiip before reading the whole instructions when encountering a question she searches the pages for the answer, finds it and moves on placing pieces on board even though she doesn't know what they do batteries as an afterthought focuses on legal connections from the start, unsure about strategy NOTE: I let her look at all tiles vs starting with them flipped over for testing purposes searches for parts based on the connections needed not the type of part just to finish the ship confused about the "smooth sides" read top to bottom left to right but kind of skimmed doesn't know what the max 5cents square on ship board is discovered universal connectors after building ship, thought they could only match with each other	Layout is not "good for eyes" - yellow is too bright, font could be better and prettier, the user really needs to focus in order to understand Game could be very immersive, could be fun but there is too much to understand - too many concepts to understand - "Who is gonna remember all this? Imagine having to open the book each time to check..." "So many rules..." User was visibly tired by the end of second task "I don't think this game would be very famous" "The journey to playing could be very frustrating." "Many elements - cabins --> connectors - -> batteries" "Can the cannons be placed anywhere? Doesn't make sense?" "Seems like it is built for someone like Sheldon Cooper" "E-normous like the rulebook is and it is just the beginning."
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		to read the rules and understand the game good to highlight important text, keeps eye focused skipping over yellow boxes because they're distracting from instructions, storyline doesn't apply to instructions, nice to have but not needed during instructions like how for double entrance of canons, it explains right into batteries vs needing to search for it later on yellow boxes provide text separator but not brain break from reading "can't move components once been added to ship" however need batteries to power different things but only for single use - confused how the game is played a lot of info on first ship compared to how is the game played? is the first round just building a ship? a lot of information without explaining the how or the why can there be a continuation of components as a heading on page 4, keep continuity, make them in a special highlighted box?	feels like a lot of wasted effort knows loose connectors are bad but not terrible- how bad? different shades of yellow are confusing- title header, boxes, emphasised rules? which ones are actually important? never actually read every word mostly legal ship, had one battery in front of a cannon "odd bits with unknown purpose" feels stupid wants a clear and concise list of the rules diagrams very helpful but layed out bad with other content on the page "smooth sides cannot connect with other connectors" is unclear, does this mean the layout is illegal or just that the connector will be loose	
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		reword "don't need to worry about this right now" - doesn't connect to building your first ship do we need to know about cabins and crews at this moment? humor helps to keep it light why would i want to go first? provide example with mistakes included - something with arrows, don't need pop quiz -having pop-quiz in FAQs vs in the first part		
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Participants 4-7 notes

"Oh my god wow" upon opening the rule book to see it

Ishan has played before

Pictures were useful (pointing to as references)

"It's very intimidating"

Teaching of movement with jets done through example and reference to physical pieces

Some passing rulebook back and forth between newer players

Forgot about whether asteroids can come from back

Some players parsing rule book while explaining

Several instances of glancing at rule book, being intimidated, and closing it

Aliens: experienced player says here's what they do, new player points to rules and says "it says here don't worry about it"

"This is board 1, right?"

"[Exasperated sigh]"

Design thought was cool, aesthetics

Non-rulebook-holding player: "I'll just follow along"

"Your ship must hold together at all times... what does that mean??"

Jumping to highlighted text

"Can you tell me" about asteroid mechanics?

"Oh well it must be here" about meteor next to picture of that on page 7 (it wasn't, had to turn the page)

Violation of minimizing memory load!

17 min to feel ready to play

Took a while to find out how to put ship on tracker in middle

Text dense, not broken up right

Needed to check rocket rules

Forgot to check each other's ships

Building took about 5 minutes

"The center counts for two, yeah I remember that"

Ayesha remembered that who finishes first goes first

"I wish you asked me sooner" about what he was reading

Format bad

Like a manual (scary)

Red X and green check with wall of text like a manual

Lasers are more effective forward than sideways, noticed that after reading

Misunderstood "two spaces behind", then like 15s later "oh, one empty space" between; others knew exactly where to go though

"Go to the combat zone" "yeah I'm looking for the adventure cards"

Thought the -3 days from combat zone was -3 crew members

Missed that you have to spend energy for engines at first, but did notice a bit later (is in different location from rest of explanation)

Went back to the "What can happen to you" in the beginning to figure out how engines work

"Maybe it's before this" (it wasn't) (engine powering for combat zone, looking all the way to beginning of rules)

Missed that sideways lasers are 0.5

Missed that all pieces fall off if partly disconnected

Missed that everyone takes damage from meteor swarm (because it is buried in wall of text, and first thing says "leader rolls")

"I'm sure there's something happening with the movement but I'm just not sure"
(didn't know when you get to move [open space])

Open space (they thought move forward, but did have to check)

Who does stardust happen to?

Did stardust leader first

Two players never noticed improper ships

Abandoned station: do I lose 5 people??

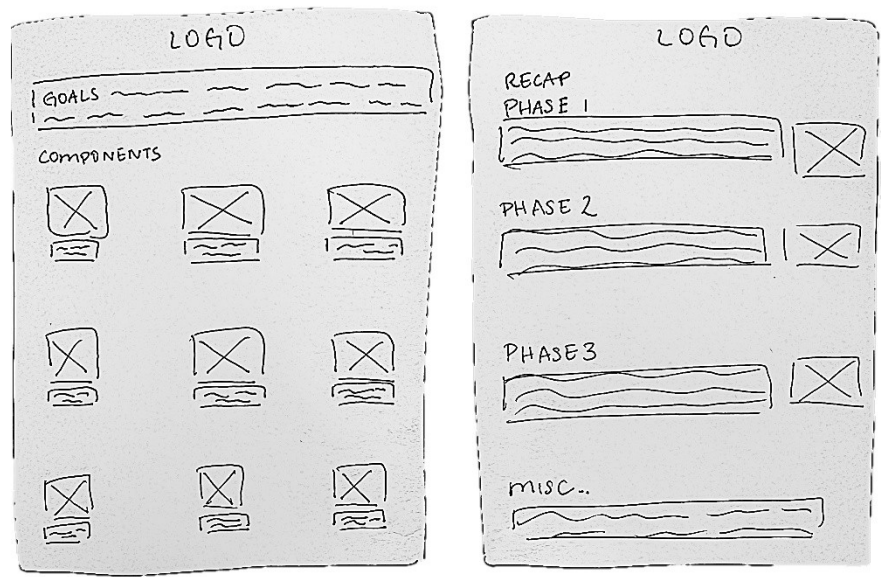
Was never paying the flight day cost in the bottom right, for, for example the abandoned station

The discarded pieces point loss was unclear, they thought max was 2 because of 2 wide

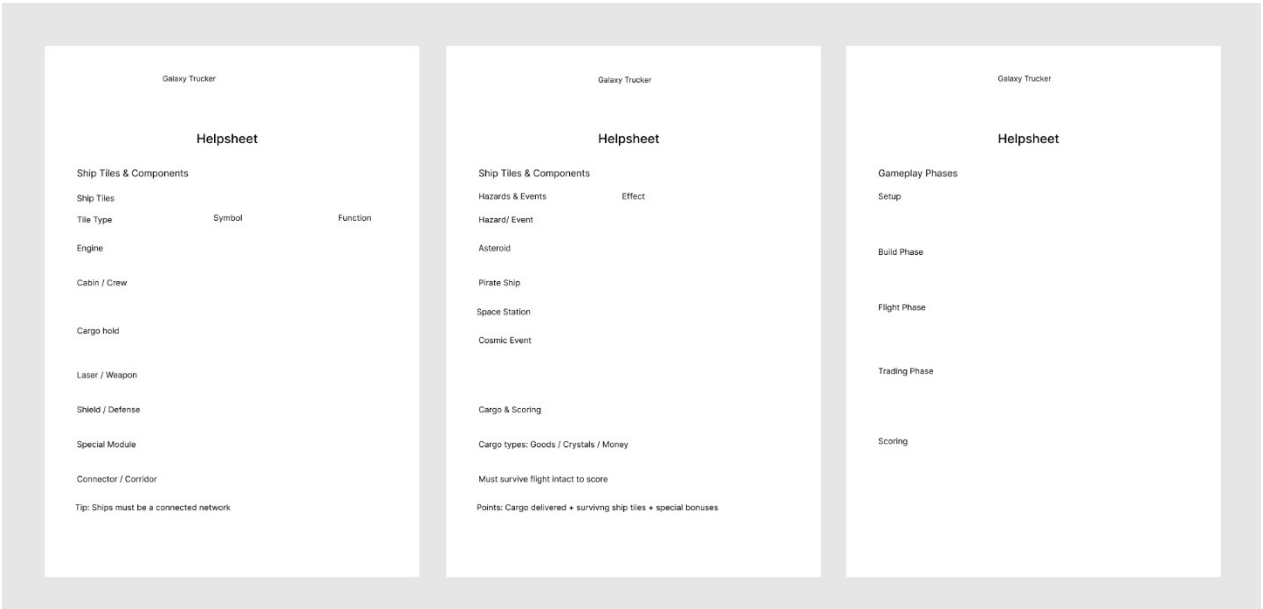
What is the cards with [+2c] on the board?

REDESIGN PAGES

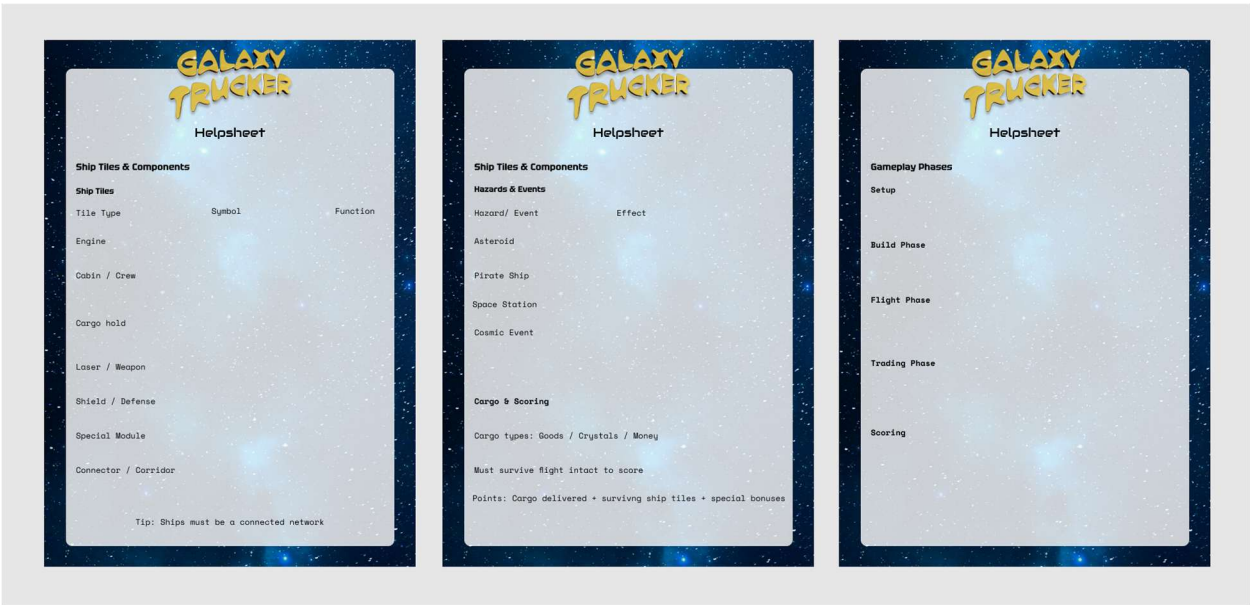
Mock-Up



Low-Fi



Mid-Fi



Hi-Fi

